Inverted MAML Card 1: Round 1 Anti-Windup & Bit-Slicing Baseline Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.03, L2=1024, Mom=0.9, Clamp=±75 LSB

1. Verification Summary

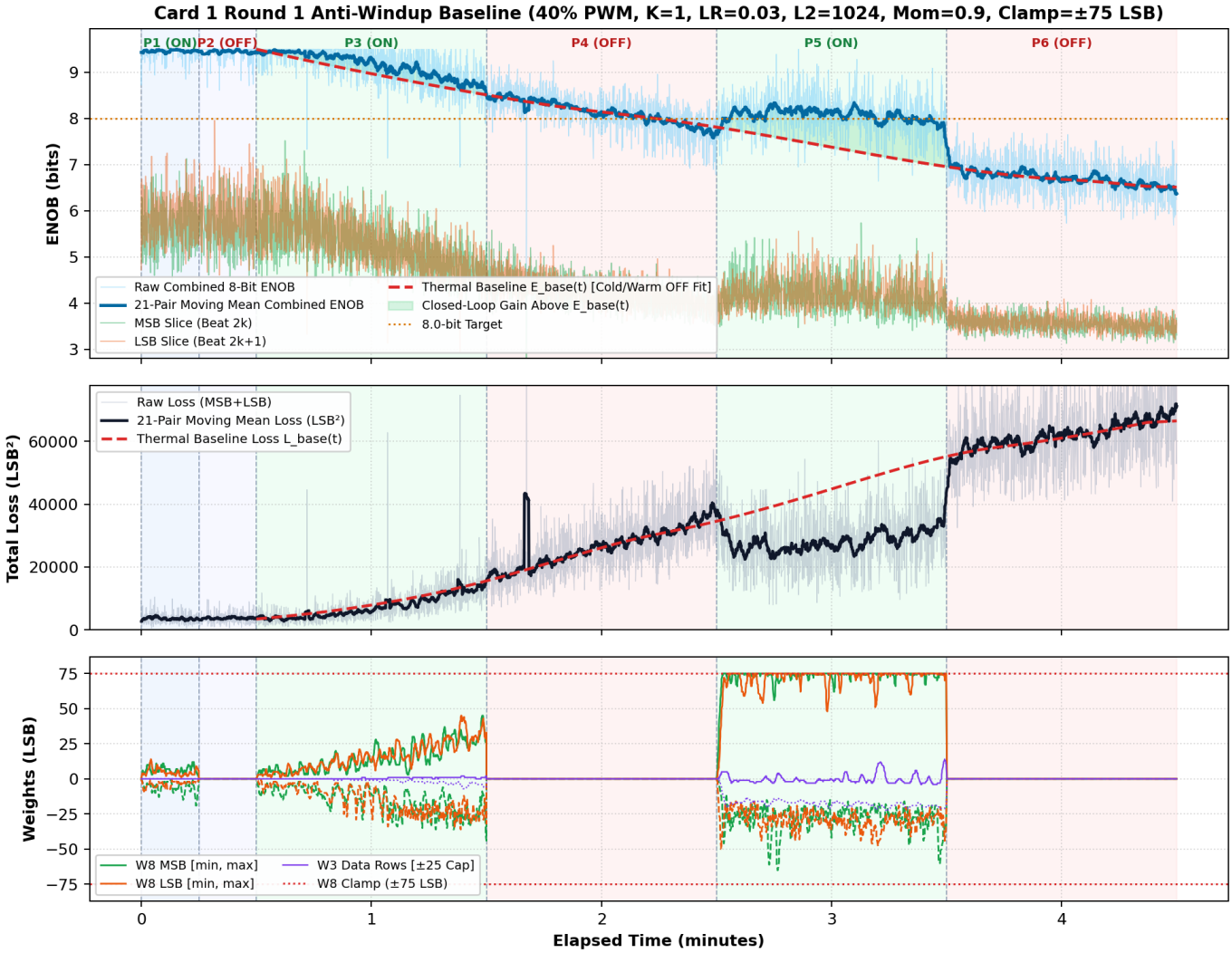
• **Cold Baseline (Heaters OFF, K=1):** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Thermal-Detrended Closed-Loop Performance:** Detrended Warm-ON Step Lift = **+0.66 b**, Loss Ratio = **1.57x**, Convergence Climb = **+0.74 b**, Anti-Windup Recovery = **0.67**, Score = **+1.957**.

• **Hot Convergence Proof (P5 vs P6):** In Warm / Regler ON (P5), ENOB climbed from **7.74 -> 7.99 b** (mean **8.07 b**) while Loss dropped (18,781 -> 16,896 MSB; 17,635 -> 13,715 LSB). Switching to Warm / Regler OFF (P6) caused immediate degradation to **6.96 -> 6.59 b** (mean **6.71 b**) with 0% W8 leak.

2. Phase-by-Phase Metrics (4,289 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	312	9.43->9.46 (9.46b)	-	3492->3478	[19,14]	0	311
P2	Cold / Regler OFF	269	9.46->9.44 (9.45b)	-	3129->3695	[0,0]	0	0
P3	Warm / Regler ON	1082	9.44->8.88 (9.17b)	+0.21 b	3970->11035	[45,45]	7	1081
P4	Warm / Regler OFF	973	8.43->7.89 (8.14b)	-0.05 b	17507->32520	[0,0]	0	0
P5	Warm / Regler ON	850	7.74->7.99 (8.07b)	+0.88 b	36416->30612	[75,75]	23	850
P6	Warm / Regler OFF	803	6.96->6.59 (6.71b)	+0.02 b	54265->65928	[0,0]	0	0


Card 1 Round 1 Anti-Windup Baseline (40% PWM, K=1, LR=0.03, L2=1024, Mom=0.9, Clamp=±75 LSB)
P1 (ON)
P2 (OFF)
P3 (ON)
P4 (OFF)
P5 (ON)
P6 (OFF)
ENOB (bits)
Raw Combined 8-Bit ENOB
21-Pair Moving Mean Combined ENOB
MSB Slice (Beat 2k)
LSB Slice (Beat 2k+1)
Thermal Baseline E_base(t) [Cold/Warm OFF Fit]
Closed-Loop Gain Above E_base(t)
8.0-bit Target
Total Loss (LSB²)
Raw Loss (MSB+LSB)
21-Pair Moving Mean Loss (LSB²)
Thermal Baseline Loss L_base(t)
Weights (LSB)
W8 MSB [min, max]
W8 LSB [min, max]
W3 Data Rows [±25 Cap]
W8 Clamp (±75 LSB)
Elapsed Time (minutes)